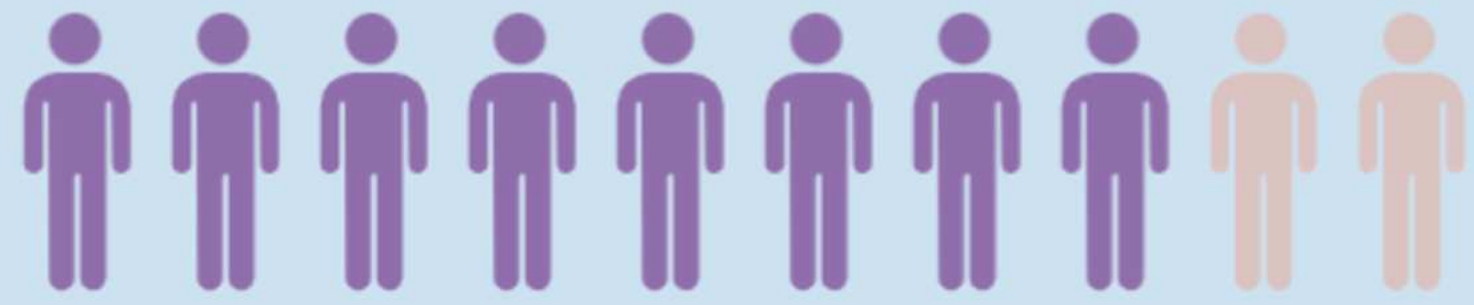


THE EFFECTS OF GLUTAMIC-PYRUVIC TRANSAMINASE-TARGETING siRNA ON HEPATOCELLULAR CARCINOMA AS MODELED IN HEPG2 CELLS

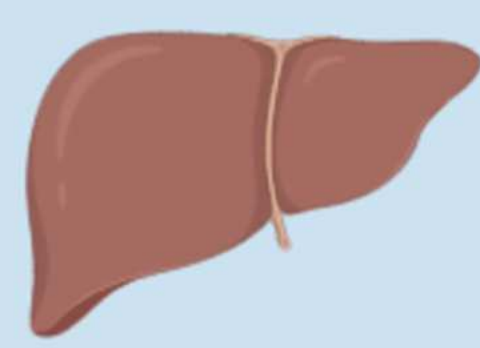
Background

Hepatocellular carcinoma (HCC) is an aggressive cancer, characterized by the **overexpression of ALT** (Guo et al., 2020)

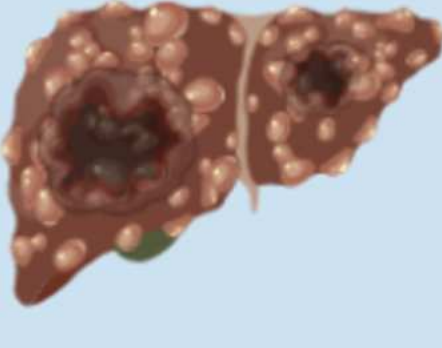


Graphic created by researchers using Biorender, 2026.

8 in 10 HCC patients die from **large tumor formation, severe liver dysfunction, and cirrhosis (scarring)** (Samant et al., 2021)



Healthy liver



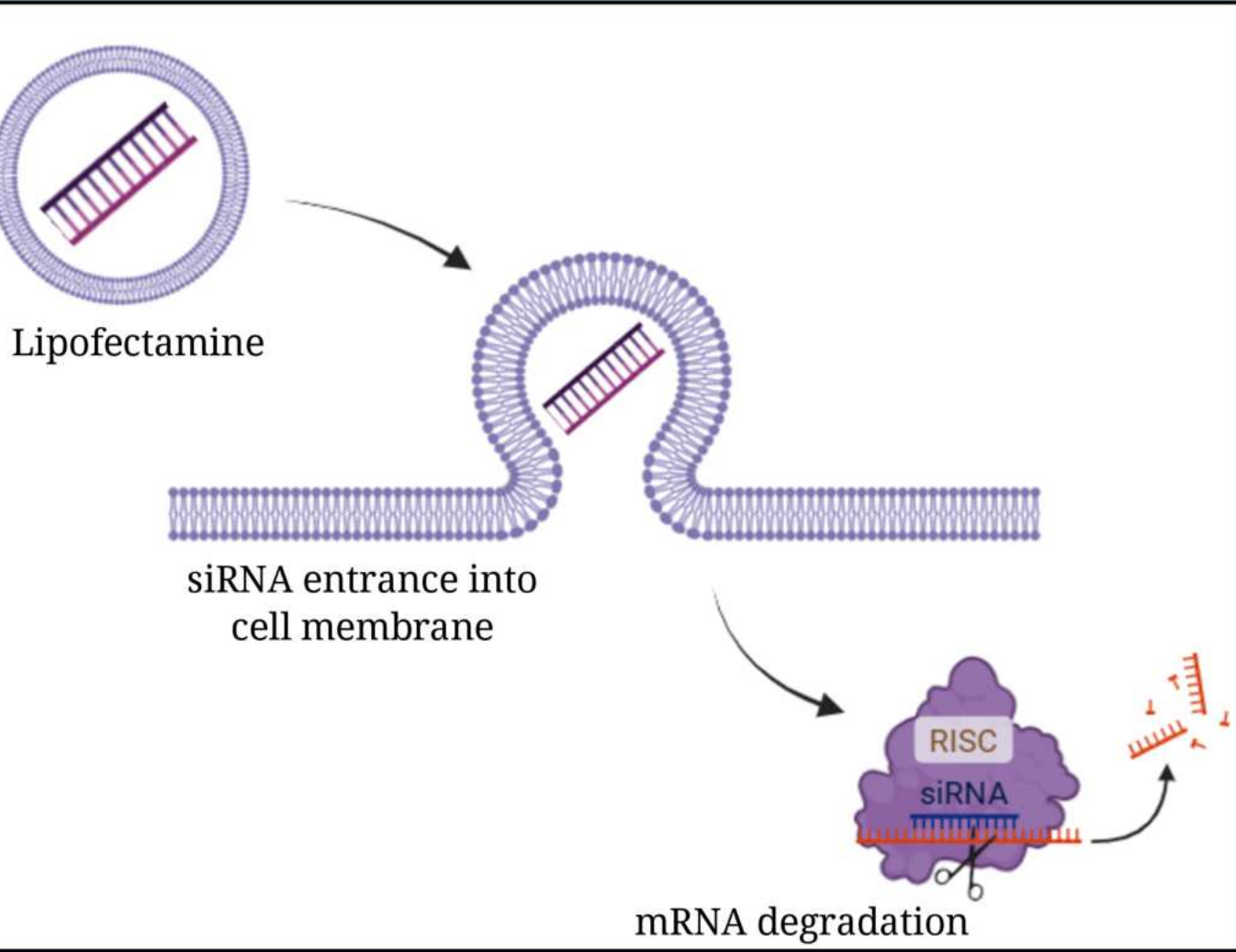
HCC-affected liver

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Overexpression
of ALT

Inflammation
and tumor
formation in
liver cells (HCC)

Alanine Aminotransferase (ALT)
is a protein found primarily in the liver coded by the **Glutamic-Pyruvic Transaminase (GPT)** gene (Fu et al., 2022)



Graphic created by researchers using Biorender, 2026.

GPT-targeting siRNA

cleaves ALT-encoding mRNA, which should **decrease ALT expression** and overall HepG2 tumor formation (Alshaer et al., 2021).

Significance

The purpose of this novel treatment is to offer an alternative to invasive and demanding current methods of treatment by **suppressing the expression of ALT**.

Hypothesis

If HepG2 cells are transfected with GPT-targeting siRNA, then **ALT protein will decrease** and **cell viability will decrease**. siRNAs silence the GPT gene, so the production of ALT will be suppressed. With these crucial pathways impacted, the cancerous cells will be less viable.

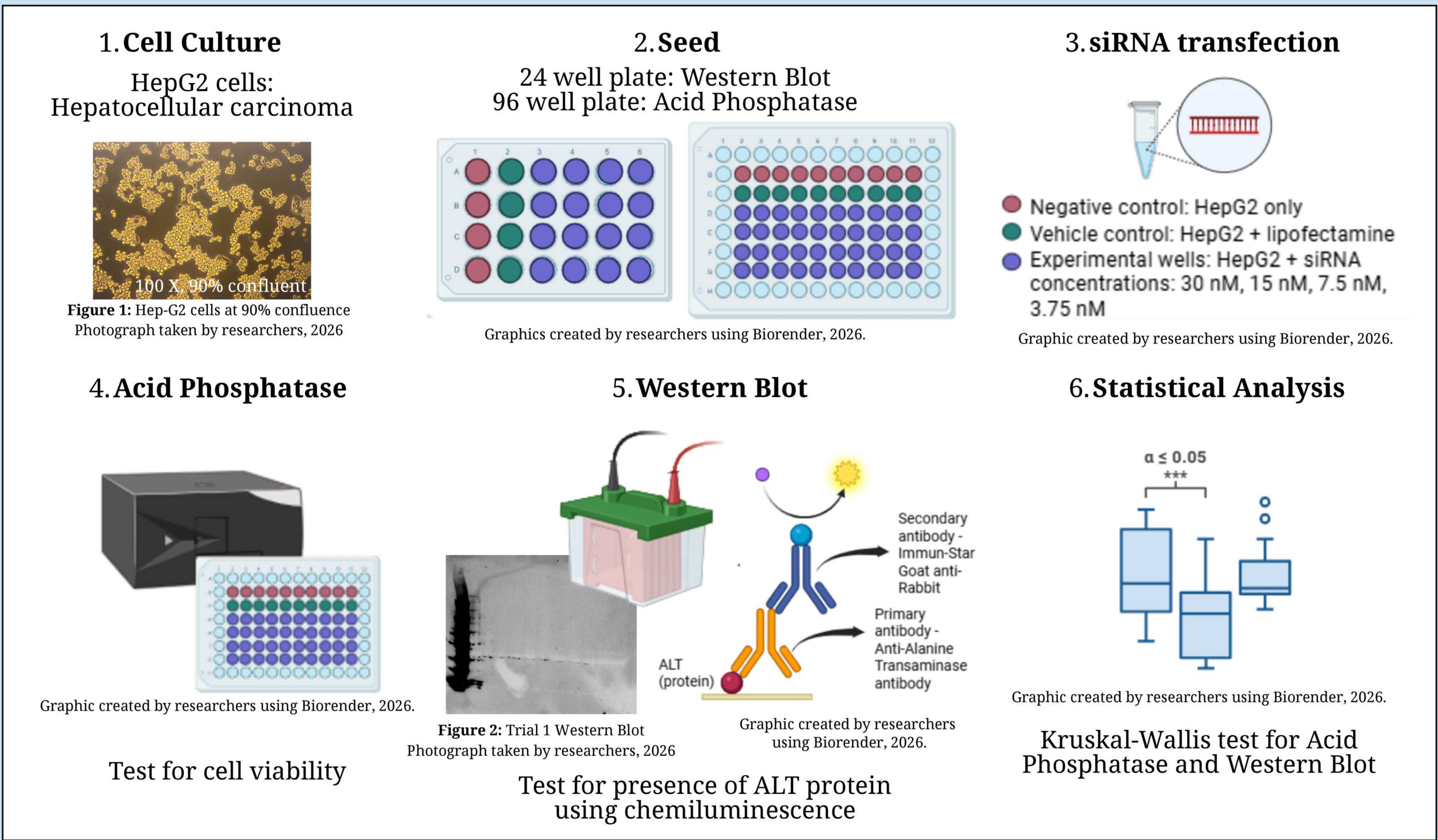
Variables

IV: GPT-targeting siRNA (30 nM, 15 nM, 7.5 nM, 3.75 nM)

DV: ALT expression & cell viability

Controls: Vehicle (lipofectamine RNAiMAX), Negative (90% EMEM, 10% FBS)

Methodology



1. Cell Culture

HepG2 cells:
Hepatocellular carcinoma

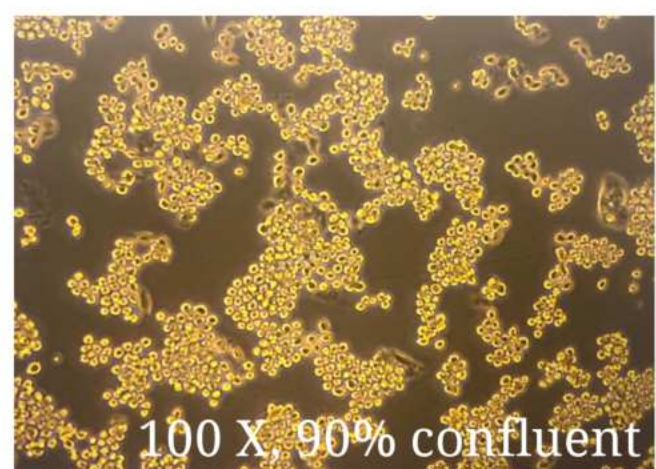
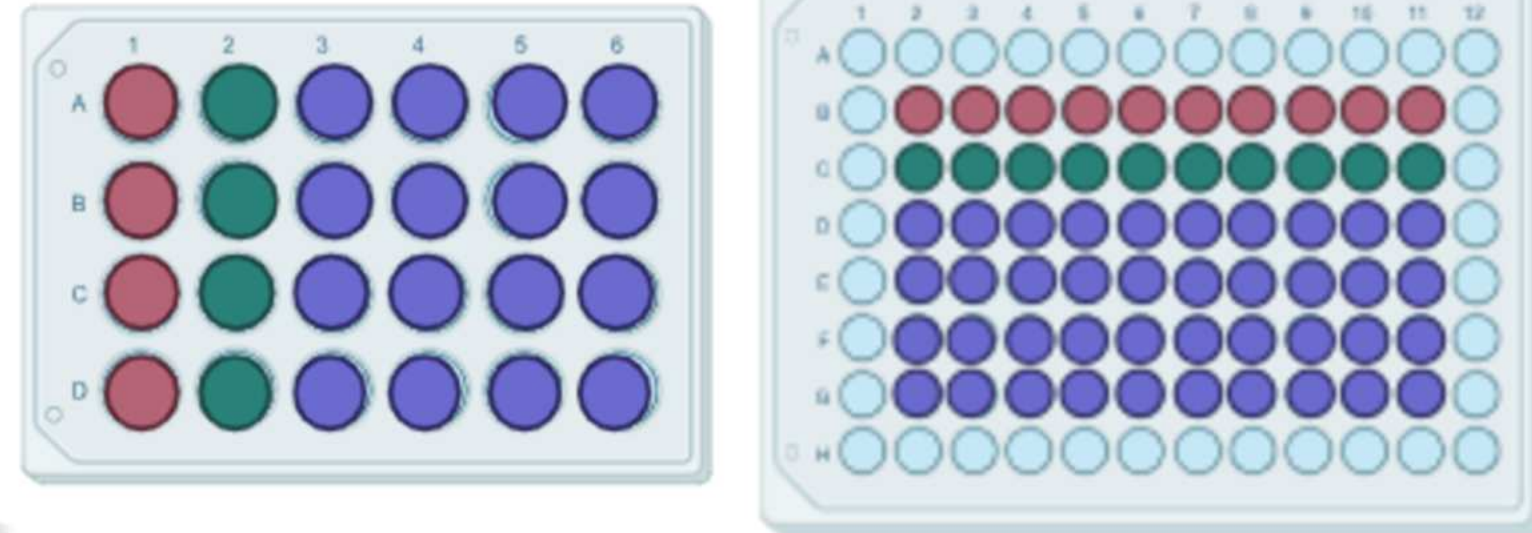


Figure 1: Hep-G2 cells at 90% confluence
Photograph taken by researchers, 2026

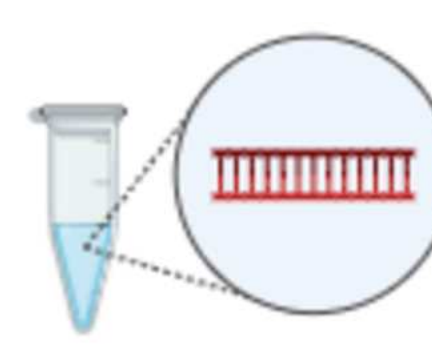
2. Seed

24 well plate: Western Blot
96 well plate: Acid Phosphatase



Graphics created by researchers using Biorender, 2026.

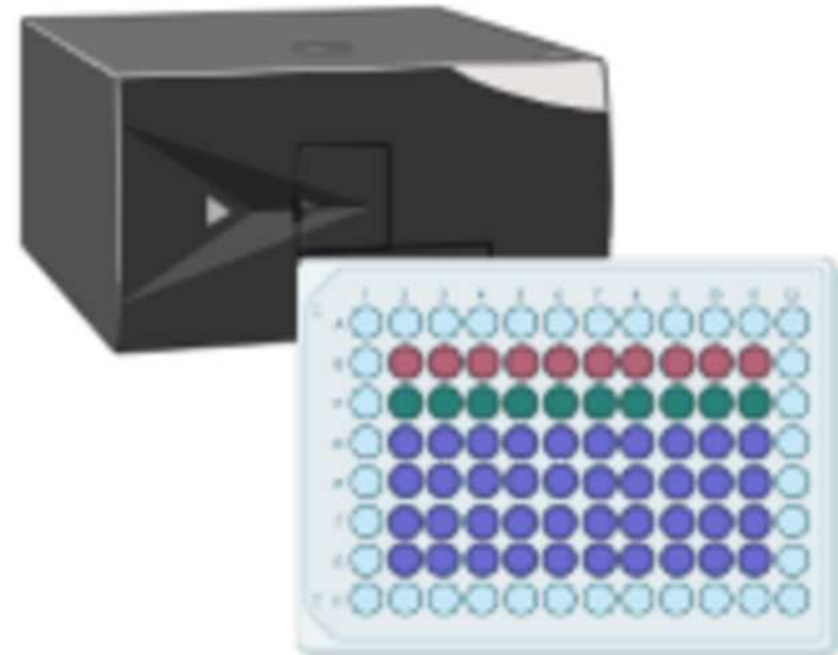
3. siRNA transfection



- Negative control: HepG2 only
- Vehicle control: HepG2 + lipofectamine
- Experimental wells: HepG2 + siRNA concentrations: 30 nM, 15 nM, 7.5 nM, 3.75 nM

Graphic created by researchers using Biorender, 2026.

4. Acid Phosphatase



Graphic created by researchers using Biorender, 2026.

5. Western Blot

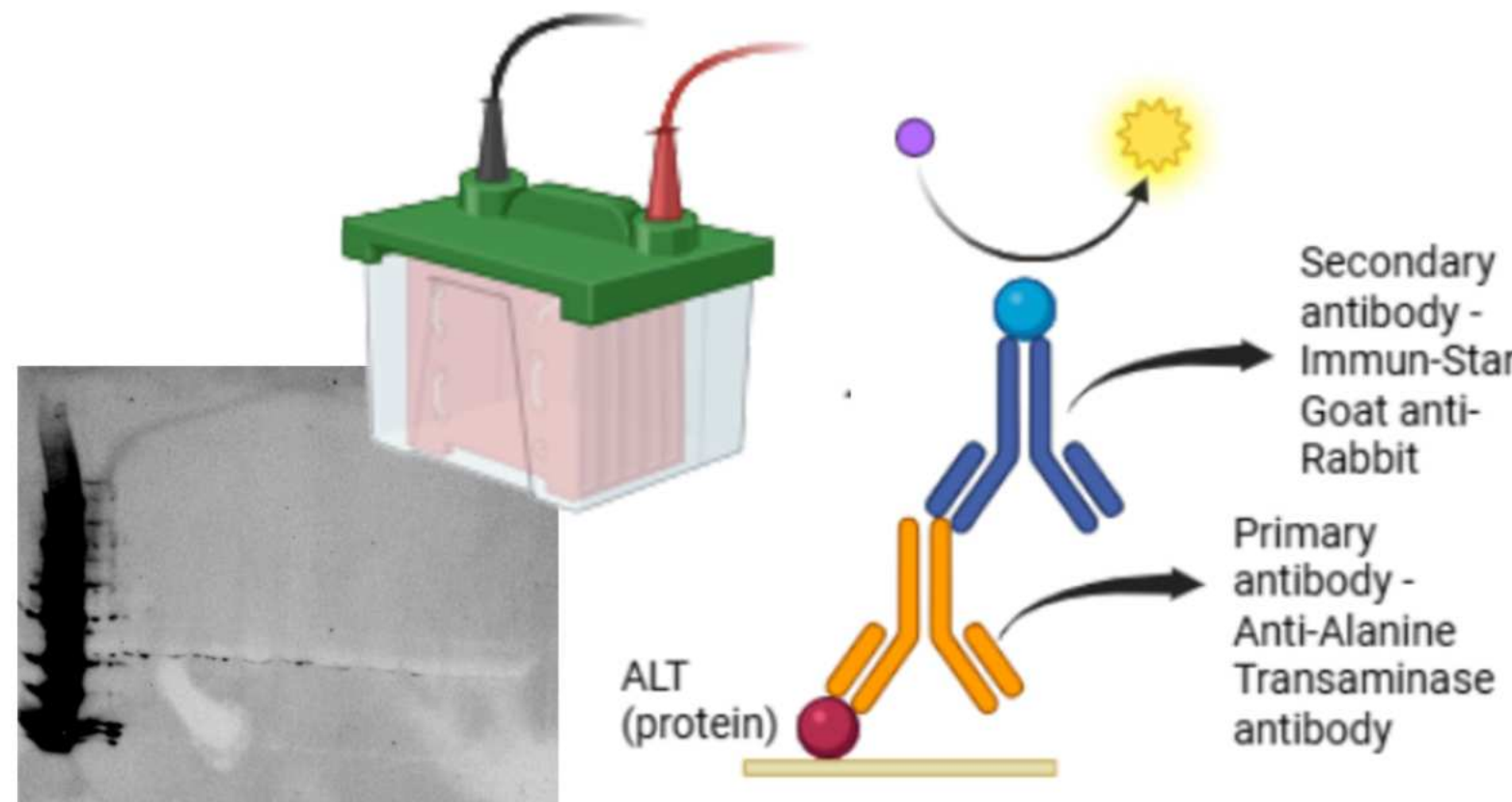
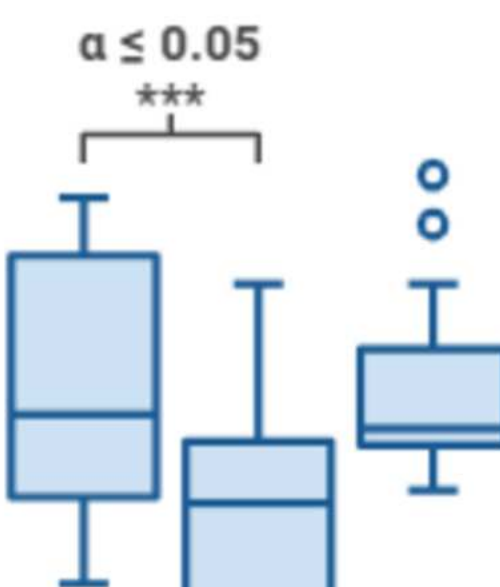


Figure 2: Trial 1 Western Blot
Photograph taken by researchers, 2026

6. Statistical Analysis



Graphic created by researchers using Biorender, 2026.

Results

Acid Phosphatase Assay

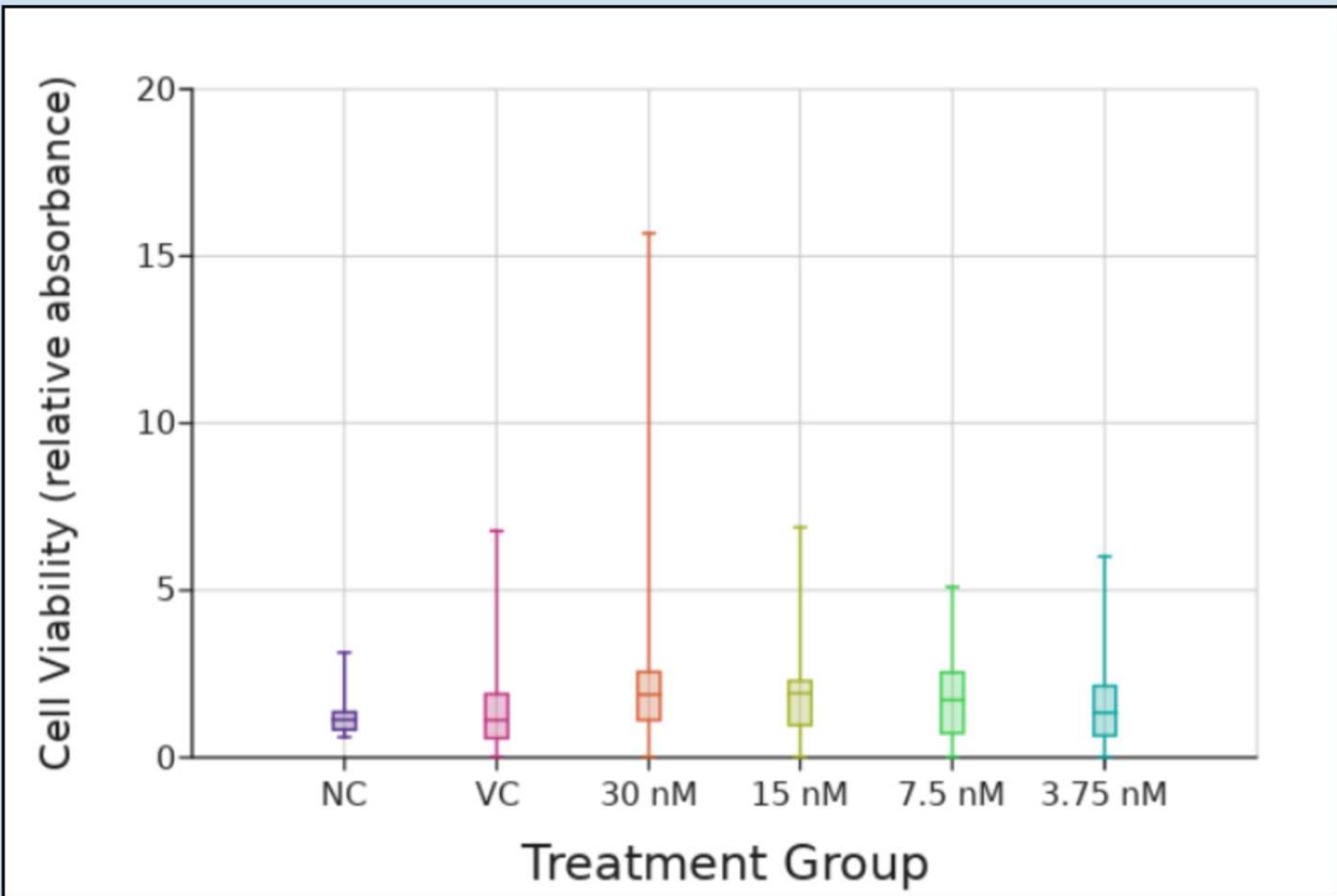


Figure 3: Median Cell Viability (relative absorbance) vs. Treatment Groups
Data Visualization created by researchers using DataClassroom, 2026.

Western Blot Assay

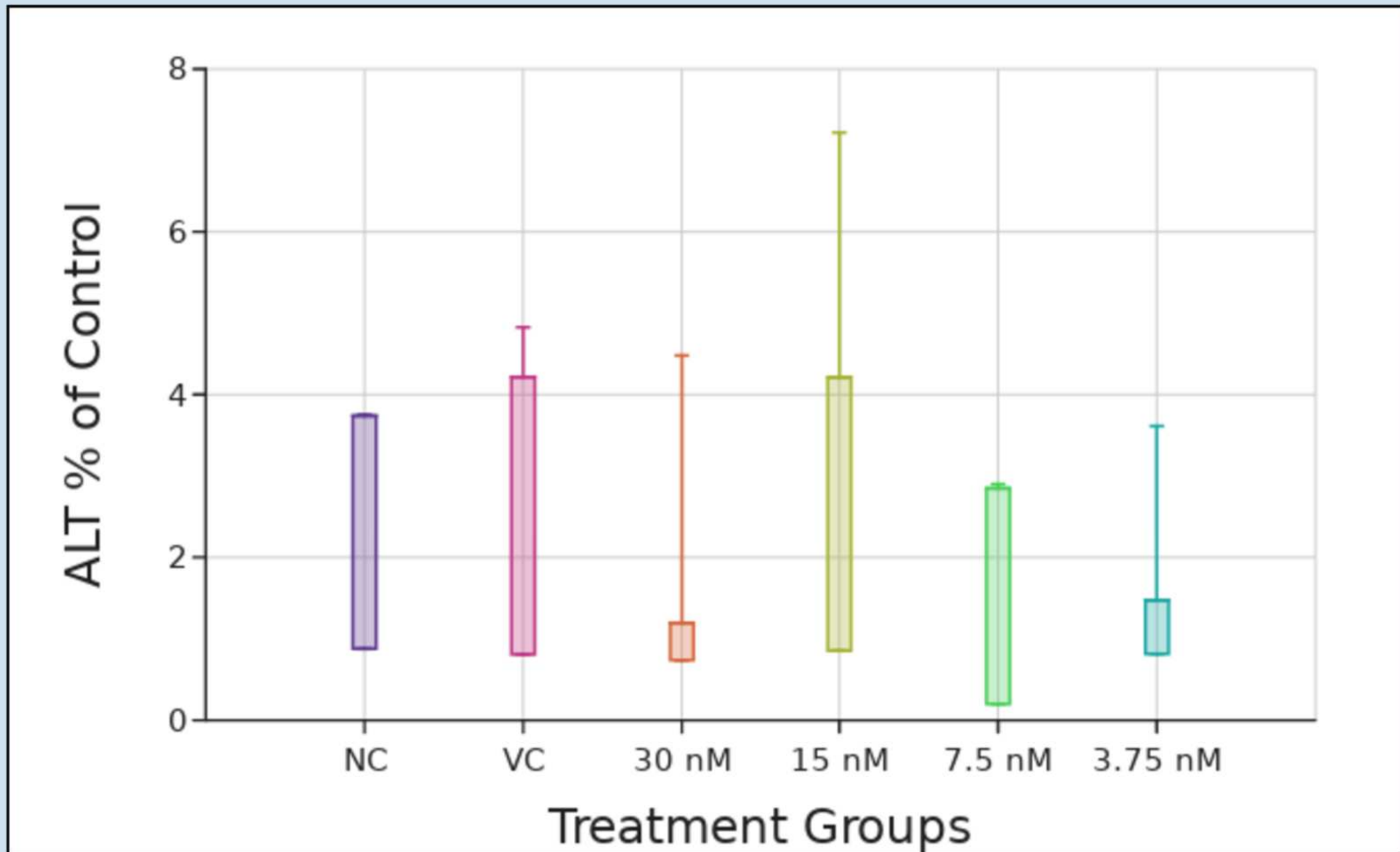


Figure 4: Median ALT % of Control vs. Treatment Groups
Data Visualization created by researchers using DataClassroom, 2026.

Statistical Analysis Results

Acid Phosphatase

Pairs with Significant Differences	P-values	Kruskal-Wallis Test P-value
NC - VC	1.00	0.07 (A P-value of 0.07 means some confidence that the groups are different)
NC - 30 nM	0.35	
NC - 15 nM	0.27	
NC - 7.5 nM	1.00	
NC - 3.75 nM	1.00	

Table 1: Kruskal-Wallis p-values comparing cell viability (relative absorbance) across the different treatment groups
Table created by researchers using Google Sheets, 2026.

Western Blot

Pairs with Significant Differences	P-values	Kruskal-Wallis Test P-value
NC - VC	1.00	0.74 (A P-value of 0.74 means no evidence that the groups are different)
NC - 30 nM	1.00	
NC - 15 nM	1.00	
NC - 7.5 nM	1.00	
NC - 3.75 nM	1.00	

Table 2: Kruskal-Wallis p-values comparing ALT expression across the different treatment groups
Table created by researchers using Google Sheets, 2026.

Discussion

Preliminary data collected thus far **did not support** hypothesis.

Statistical analysis concluded that there was **not enough evidence to prove a significant difference** between cell viability and ALT protein expression of different treatment groups. More trials should be conducted.

Acid Phosphatase

- **No statistically significant difference** in cell viability across siRNA concentrations
 - Minor fluctuations in cell viability were observed when comparing treatment groups, but overall remained within the range for viable HepG2 cells
- **No statistically significant difference** in cell viability between negative control (NC) and vehicle control (VC)
 - **Supports validity** of utilizing lipofectamine RNAiMAX as siRNA transfection reagent

Conclusion:

GPT-targeting siRNA did not significantly decrease HepG2 cell viability in current experimental conditions

Western Blot

- **No statistically significant difference** in ALT protein levels (shown as ALT % of control) amongst negative control (NC) and siRNA concentrations
 - Quantification **did not reveal a decrease in ALT** presence consistent with the concentration curve
- ALT protein expression remained relatively stable despite siRNA transfection and treatment

Conclusion:

GPT-targeting siRNA did not produce an identifiable decrease in ALT protein expression in current experimental conditions

Implications

- ALT-targeting approaches need **more optimized or stronger gene-silencing methods**
- HCC treatment may require **targeting various pathways** due to complex metabolism
- siRNA-based treatments offer promising **therapeutic benefit**

Future Research

- Conduct **more trials** to compare to preliminary data
- Conduct additional trials utilizing diverse transfection methods to **improve gene knockdown efficiency**
- Use qPCR Assay to confirm GPT knockdown and **measure GPT mRNA reduction** directly
- Compare results of HepG2 cells with **other liver cancer models**

References

Alshaer, W., Zureigat, H., Al Karaki, A., Al-Kadash, A., Gharaibeh, L., Hatmal, M. M., Aljabali, A., & Awidi, A. (2021). siRNA: Mechanism of action, challenges, and therapeutic approaches. *European Journal of Pharmacology*, 905(1), 174178. <https://doi.org/10.1016/j.ejphar.2021.174178>

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Guo, W., Tan, H.-Y., Li, S., Wang, N., & Feng, Y. (2020). Glutamic-Pyruvic transaminase 1 facilitates alternative fuels for hepatocellular carcinoma growth—a small molecule inhibitor, berberine. *Cancers*, 12(7), 1854. <https://doi.org/10.3390/cancers12071854>

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